

Mawlana Bhashani Science and Technology University

Department of Information and Communication Technology



SSH
(Secure Shell)

Data Communication Lab · ICT-3102

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Presentation Outline

01 Introduction: why is it called Secure Shell?

02 Related work: from Telnet to SSH

03 How SSH works and public-key cryptography

04 SSH in practice, features and comparison

05 Conclusions and best practices

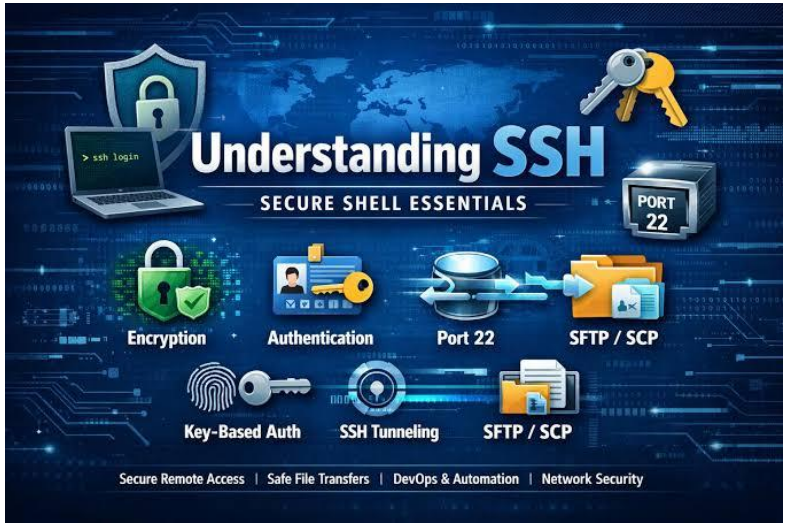
Introduction: Why “Secure Shell”?

SSH is called Secure Shell because all communication between client and host is encrypted.

01 Telnet sends passwords and commands in plain text

02 Anyone on the network can sniff and steal credentials

03 Goal: see how SSH secures remote access



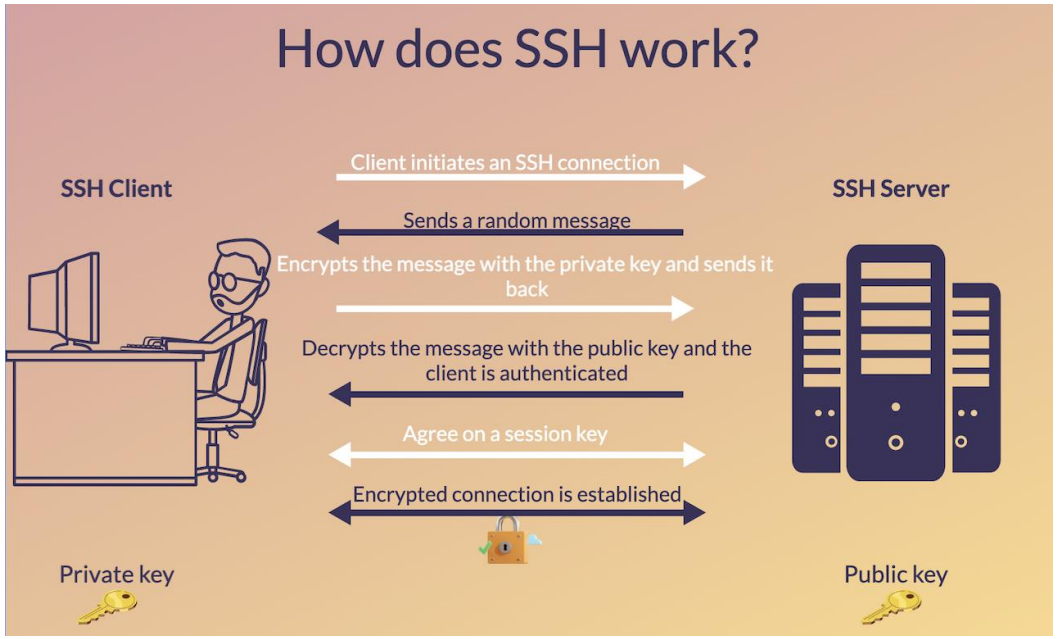
Related Work and Research Gap

Study / approach	Key idea	Relevant limitation
Telnet / rlogin / rsh	Remote login and commands	Plain-text passwords, easy to sniff
SSH-1 (Ylönen, 1995)	First encrypted replacement	Cryptographic weaknesses; superseded
SSH-2 (current)	Layered, modular protocol	MITM if host key is not verified

Research gap Legacy tools sent data in plain text; a secure, authenticated channel was needed.

How SSH Works

Authentication and encryption of an SSH connection

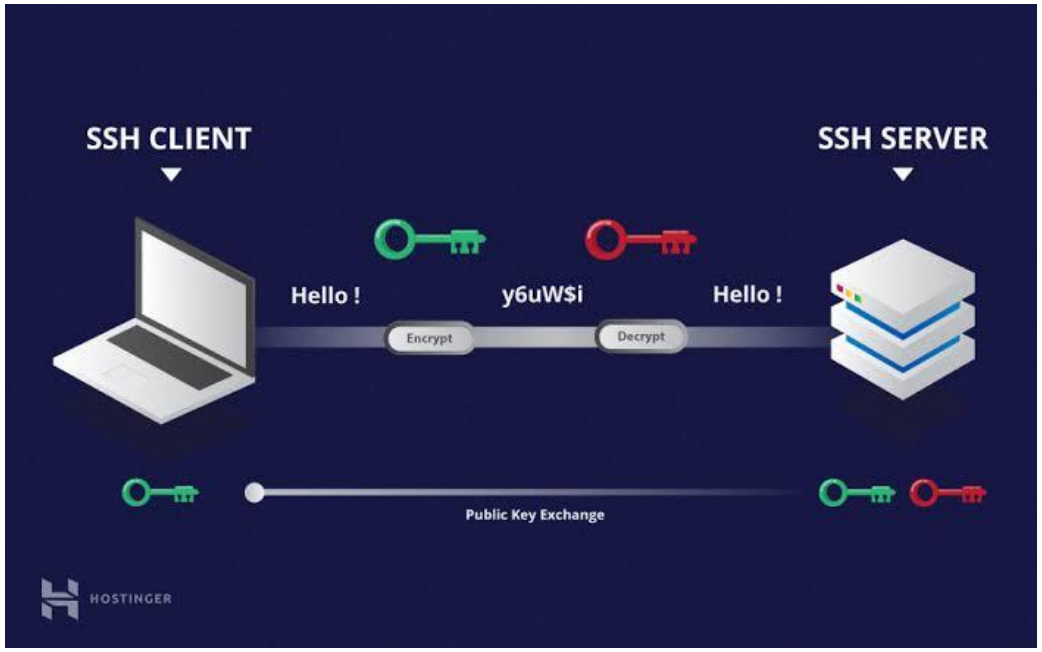


01 **Connect and verify server**
Client connects; host key checked

02 **Authenticate the user**
Private key signs a challenge

03 **Encrypted session**
Both sides agree on a session key

Public-Key Cryptography in SSH



Key idea

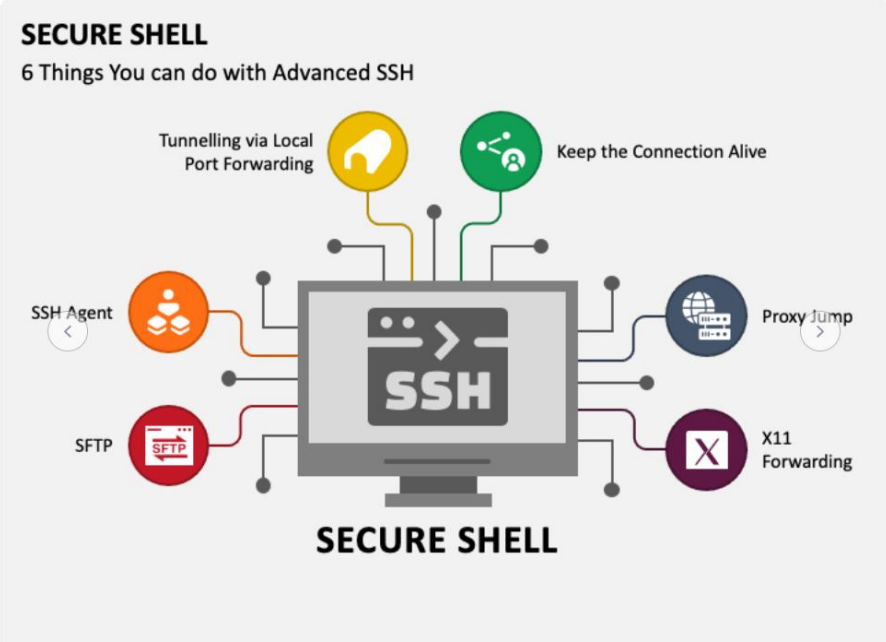
A public key and a private key form a pair.
Data locked with one key opens only with its partner.

Public-key exchange between SSH client and server (source: Hostinger)

SSH Setup and Configuration

Default port	TCP port 22; can be changed to a port above 1023
Protocol layers	Transport, User Authentication, Connection
Configuration files	/etc/ssh/: sshd_config, ssh_config, ssh_known_hosts
Implementations	OpenSSH, SSH Tectia, PuTTY, MindTerm

Advanced SSH Features



Key features

Strong encryption
Strong authentication
Authorization
Integrity of communication
Forwarding / tunneling

Six advanced capabilities: tunneling, SSH agent, SFTP, proxy jump, X11, keep-alive

SSH vs Telnet



Main finding

SSH encrypts everything; Telnet sends plain text.

Use SSH for all remote access.

SSH (TCP 22, encrypted) versus Telnet (TCP 23, plain text)

Protocol Comparison

Protocol	Encryption	Default port	Authentication
Telnet	None (plain text)	23	Password only
SSH-1	Weak, deprecated	22	Password / keys
SSH-2	Strong encryption	22	Password / public key

SSH keeps Telnet's remote-shell function but adds encryption, authentication and integrity.

Conclusions and Best Practices

SSH is the secure standard for remote access, encrypting every session.

01 Provides encryption, authentication, integrity and tunneling

02 Limitation: man-in-the-middle if the host key is not verified

03 Best practice: key-based login, verify host keys, never use Telnet

Selected References

[1] T. Ylönen, C. Lonvick, “The Secure Shell (SSH) Protocol Architecture,” RFC 4251, 2006.

[2] T. Ylönen, C. Lonvick, “The Secure Shell (SSH) Authentication Protocol,” RFC 4252, 2006.

[3] T. Ylönen, C. Lonvick, “The Secure Shell (SSH) Transport Layer Protocol,” RFC 4253, 2006.

[4] J. Postel, J. Reynolds, “Telnet Protocol Specification,” RFC 854, 1983.

Thank you

Questions and Discussion

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Any questions?